

Multi-Objective Algorithms for Truck-Drone Cooperative Delivery Routing

A comparative study of NSGA-II, PACO, PACO+ALNS and Pure ALNS on the Solomon benchmark and WCCI-2020 E-CVRP

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1 Abstract

We study the truck-drone routing problem with time windows (TDRPTW) as a bi-objective optimization problem: minimize total transportation cost and weighted tardiness.

Four algorithms (NSGA-II, improved PACO, PACO+ALNS, Pure ALNS ablation) are compared under one unified model on 54 Solomon instances x 25/50/100 x two endurance levels = 324 configurations, plus 24 E-CVRP instances.

PACO+ALNS attains the best cost rank (1.31) and HV rank (1.09), feasible in all 324 configurations.

2 Introduction & Related Work

In truck-drone cooperative delivery, each truck carries a drone that can be launched from the moving truck, serve one customer, and return to the same truck. This extends the VRPTW by service-mode assignment and temporal synchronization.

The two objectives, total transportation cost and weighted tardiness, conflict; a bi-objective formulation avoids tuning a penalty weight, and hypervolume measures Pareto-front quality.

Contributions

- all 54 Solomon instances (C17, R23, RC14), sizes 25/50/100, endurance 4/6 km = 324 configurations.
- four algorithms under one model and protocol, with grouped statistics by family, size, and endurance.
- PACO+ALNS hybrid with capacity-repair engineering (30-62% runtime reduction).
- Pure ALNS ablation and WCCI-2020 E-CVRP generalization.

Related work: Solomon (1987); Das et al. (2021); NSGA-II (Deb et al., 2002); ACO (Dorigo et al., 1996); ALNS (Ropke & Pisinger, 2006); E-CVRP (Mavrovouniotis et al., 2020).

3 Problem Formulation

Notation. Depot = node 0; customers N. Customer i has demand q_i , window $[e_i, l_i]$, priority p_i ; trucks capacity Q; drones capacity Q^d , endurance R^d .

Objective 1: total cost

$$\min C = \sum_{r \in K} (F^t + c^t \cdot L_r^{\text{truck}}) + \sum_{d \in D} (F^d \cdot z_d + c^d \cdot L_d^{\text{drone}})$$

Objective 2: weighted tardiness

$$\min T = \sum_{i \in N} p_i \cdot \max(0, a_i - l_i)$$

Constraints

$$\sum_{j \in r} q_j \leq Q \quad (\forall r \in K); \quad \sum_{j \in m} q_j \leq Q^d \quad (\forall m \in D)$$

$$d(i,j) + d(j,k) \leq R^d \quad (\forall \text{ drone mission})$$

$$w_i = \max(0, e_i - a_i); \quad t_i = \max(0, a_i - l_i) \cdot p_i$$

$$a^{\text{drone}}(k) \leq a^{\text{truck}}(k) + t^{\text{ret}}, \quad \text{violation} \rightarrow 50 \cdot p_i$$

$$n_{\text{trucks}} = \max(2, \text{ceil}(n/10)) \text{ [type-1]}, \max(2, \text{ceil}(n/25)) \text{ [type-2]}; \quad n_{\text{drones}} = n_{\text{trucks}}$$

Objective 1: F^t/c^t are truck fixed cost and cost/km; L^{truck} is the truck route length; F^d/c^d are drone fixed/variable cost; z_d indicates drone use; L^{drone} is drone flight distance.

Objective 2: a_i is arrival time; l_i is window end; p_i is priority; $\max(0, a_i - l_i)$ is the tardiness.

Constraints: q_j is demand; Q / Q^d are capacities; R^d is endurance; w_i is waiting; t_i is tardiness $a^{\text{drone}}(k) / a^{\text{truck}}(k)$ are retrieval arrivals; t^{ret} is retrieval time; violation adds $50 \times p_i$.

Fleet: **type-1** assumes about 10 customers per truck; **type-2** about 25; $n_{\text{drones}} = n_{\text{trucks}}$ (1:1 pairing).

Single depot: node 0 is the only depot; every truck route starts and ends there, and the return leg is included in cost/time.

Time windows: early arrival waits; launch and retrieval each take 0.5 min; the truck never waits for the drone.

Penalties: missing $\times 10,000$; overload $\times 1,000$; reported Pareto fronts are feasible-only.

Enforcement: NSGA-II penalizes infeasible individuals; PACO/PACO+ALNS construct feasible actions with repair; Pure ALNS checks feasibility during insertion.

4 Experimental Design & Parameters

Parameter	Value
Customers	25 / 50 / 100
Drone endurance	4 km / 6 km
Speeds	truck 25 km/h; drone 50 km/h
Capacities	truck 350; drone 40
Costs	truck 100 + 2/km; drone 0 + 1/km
Priorities	type-1 U(0.8,2.0); type-2 U(0.4,1.0)
Repeats / iterations	3 runs x 100
NSGA-II population	100
PACO ants	30
W8 ants	20-40 adaptive
W8 ALNS iterations	15-25 adaptive
Archive / decay / weights	$\max(10, \min(100, 0.25n)) / 0.85 / 10,5$

Scale: 54 Solomon instances (C17/R23/RC14) x 25/50/100 x 4/6 km = 324 configurations; E-CVRP 24 instances (E7/F3/M4/X10); budgets 30x3 / 10x2 / 5x1; F-n140 uses 7 vehicles.

Metrics: mean cost, mean tardiness, HV (shared reference), runtime, feasibility; feasible-only fronts.

5 Algorithms & Penalty Handling

NSGA-II. Two-layer encoding (permutation + mode); PMX crossover; non-dominated sorting + crowding distance; truck-first decoding. Fitness adds overload $\times 1,000$ and drone-failure $\times 50$; final front feasible-only.

PACO. Dual pheromone for truck arcs and drone triples; weighted transition; archive + crowding pruning. Missing $\times 10,000$ added to cost and tardiness; archive feasible-only.

PACO+ALNS. Construction + 2-opt + relocation + ALNS on elites; 5 destroy / 3 repair operators; adaptive weights; score cost + tard $\times 10$ with missing/overload guards; capacity repair cuts runtime 30-62%.

Pure ALNS. No ACO construction/pheromone; same operators, acceptance and budget; archive adds missing $\times 10,000$ and overload $\times 1,000$; feasible-only.

Constraint enforcement. Model constraints are identical; NSGA-II penalizes infeasible individuals, PACO/PACO+ALNS construct feasible actions with repair, Pure ALNS checks feasibility during insertion; E-CVRP keeps only capacity constraints.

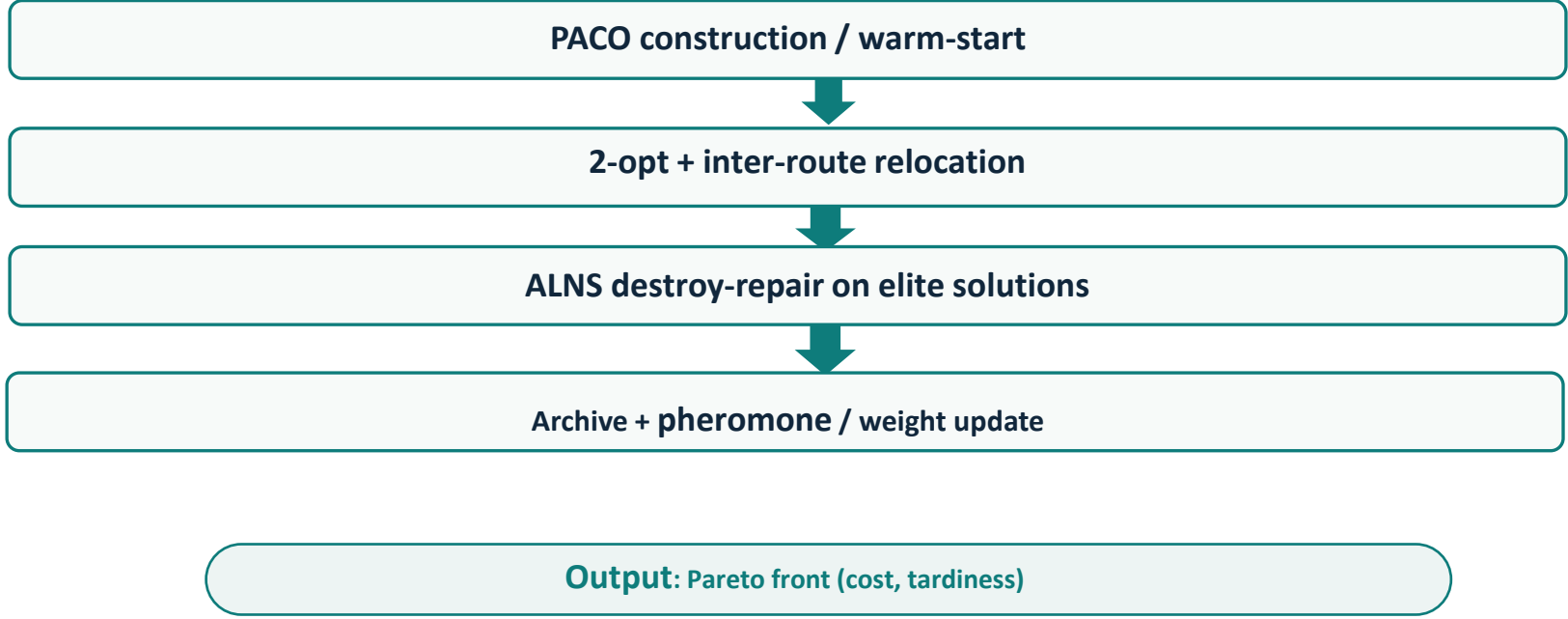


Fig. 1 PACO+ALNS pipeline.

6 Results & Analysis

Algorithm	Cost rank	HV rank	Feasible	Time (s)
NSGA-II	3.98	3.74	311/324	15.9
PACO	1.71	2.00	324/324	275.6
PACO+ALNS	1.31	1.09	324/324	467.2
Pure ALNS	2.98	3.15	318/324	29.7

Table 1. Aggregate Solomon results (feasible-only). PACO+ALNS has the best cost/HV ranks; NSGA-II wins tardiness in 206/324 cases but fails in 13 wide-window 100-customer cases.

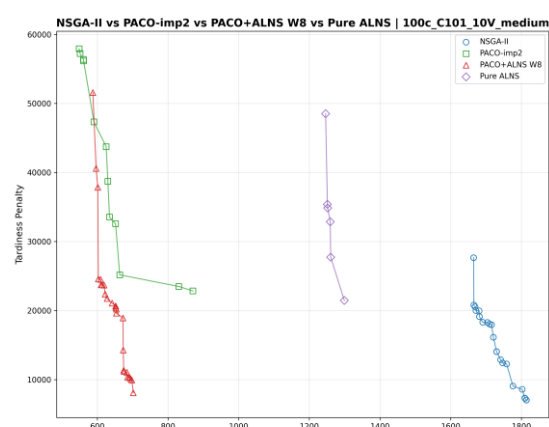


Fig. 2 C101-100c

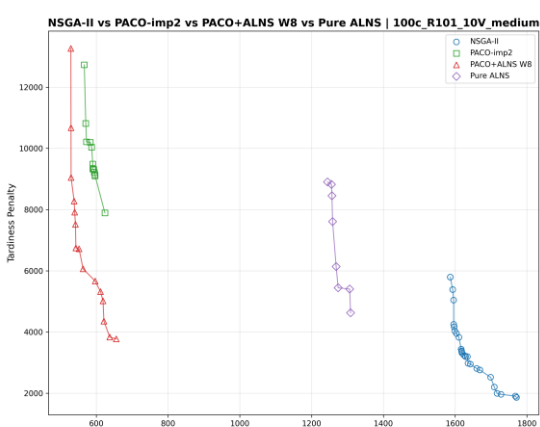


Fig. 3 R101-100c

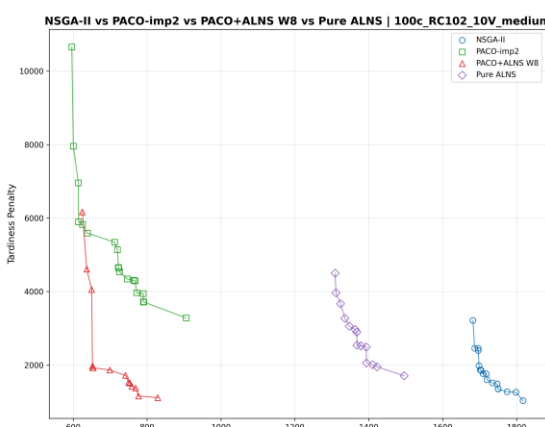


Fig. 4 RC102-100c

Fig. 2-4 interpretation. PACO+ALNS and PACO occupy the lower-cost region; NSGA-II extends along the lower-tardiness axis; Pure ALNS lies in the higher-cost region with limited coverage. PACO+ALNS gives the widest front and the highest HV in most configurations.

Grouped analysis

- lowest median cost in all C/R/RC families and at 25c/50c; at 100c PACO is slightly lower but W8 has more HV best counts (91 vs 17).
- ranks are nearly unchanged under medium/high endurance; results are robust.
- HV: W8 highest mean HV at 25c/50c; Pure ALNS narrow because of scalarization.
- runtime: W8 467.2 s, PACO 275.6 s, Pure 29.7 s, NSGA-II 15.9 s.
- feasibility: PACO and W8 324/324; NSGA-II 311/324; Pure 318/324.

Table 2: E-CVRP generalization (CVRP relaxation).

Algorithm	Mean gap	Feasible
NSGA-II	+67.44%	21/24
PACO	+9.63%	22/24
PACO+ALNS	-1.01%	24/24
Pure ALNS	+3.32%	24/24

Explanation of -1.01%: Published optima include energy/recharging constraints; our CVRP relaxation removes them, so a lower distance is attainable. The gap refers to the relaxed problem, not the true EVRP optimum.

7 Discussion, Conclusions & Future Work

- trade-off: NSGA-II favors tardiness; PACO favors cost; PACO+ALNS balances both.
- ablation: Pure ALNS ranks 2.98/3.15, so ACO construction + pheromone are key.
- generalization: PACO+ALNS is -1.01% on E-CVRP and feasible 24/24.

Conclusions: PACO+ALNS provides the best cost-HV trade-off and robust feasibility.

Future work: equal-time budgets, energy constraints, significance tests, RL-guided operators, multi-depot/dynamic demand.

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